CNN PROJECT • SUBMISSION 1 OF 4

Attention-Guided Fine-Grained Classification of Cricket Batting Shots from Single Frames

Dataset, Exploratory Analysis and Related Work

Repository: github.com/Rayhan-who/cricket-shot-attention-cnn

Summary

This submission establishes the dataset and evaluation protocol for a four-part CNN project on cricket batting shot recognition. The task is **multi-class, single-label image classification** over four shot types, performed on **individual still frames** rather than video sequences.

Three outcomes are delivered here. First, a complete inventory and exploratory analysis of the dataset. Second, a **duplicate audit** establishing that the published dataset contains offline-augmented copies of the same source photographs, and a correspondingly leakage-safe train/validation/test split that all later submissions reuse unchanged. Third, a survey of prior work establishing that the field is almost entirely video-based, that published accuracies in it are known to be difficult to reproduce, and that the contribution of attention has not previously been isolated.

> **Framing.** Almost all prior cricket shot classification work models temporal sequences with LSTM or GRU layers over multiple frames. This project removes that signal entirely and asks how much of the task is recoverable from a single frame — and whether spatial attention can close part of the resulting gap.

1 Dataset information

1.1 Provenance

> **Action required.** The two fields marked ____ below must be read directly from the dataset's Kaggle page and entered before final submission. If the licence field reads "Unspecified", write *Unspecified* — do not infer a licence.

Field	Value
Name	Cricket Shot Dataset
Source	kaggle.com/datasets/aneesh10/cricket-shot-dataset
Author	Aneesh (Kaggle user `aneesh10`)
Licence	___
Year published	___
Task type	Multi-class, single-label image classification
Number of classes	4
Class names	`drive`, `legglance-flick`, `pullshot`, `sweep`
Construction	Images collected from internet sources, then expanded by offline augmentation

1.2 Class definitions

Class	Shots included
`drive`	Cover drive, straight drive, off drive
`legglance-flick`	Leg glance and flick
`pullshot`	Pull
`sweep`	Sweep

1.3 Measured properties

The following are computed automatically by the accompanying notebook (`01_eda_and_related_work.ipynb`) and must be transcribed from its output.

Property	Measured value
Total images	___
Images per class	___
Image dimensions (min / median / max)	___
Distinct size combinations	___
Colour channels	___
File format	___
Total size on disk	___
Corrupt or unreadable files	___

1.4 Why this dataset matters in its domain

Cricket has an audience exceeding one billion people, and the broadcast and coaching workflows built around it remain substantially manual. Automatic batting-shot recognition supports three applications that recur throughout the literature:

- **Highlight generation.** Broadcasters currently tag shot types by hand across many hours of footage. Automatic labelling is a direct cost reduction.
- **Coaching and technique analysis.** Shot-type identification is the precondition for any downstream biomechanical feedback — a system cannot assess the execution of a pull shot until it knows a pull shot was attempted.
- **Automated commentary and context-aware advertising.** Both require machine-readable understanding of what is happening on the field.

What makes this a genuinely difficult classification problem, rather than a trivial one, is that all four classes share an identical scene grammar: one batsman, one pitch, the same protective equipment, comparable stadium backdrops, and frequently a crowd and sightscreen in the background. The class label is determined almost entirely by **body posture and bat angle**, which occupy a small fraction of the total frame area. Prior work makes precisely this observation — the visual similarity between batting shots is what renders hand-crafted feature extraction impractical and motivates learned representations [1].

This property is also what makes the dataset suitable for the later submissions. A coarse or visually separable dataset would saturate near ceiling accuracy under any reasonable pretrained backbone, leaving no measurable headroom for the attention module introduced in Submission 4 to demonstrate an effect. Fine-grained difficulty is a requirement of the experimental design, not an inconvenience.

2 Exploratory data analysis

2.1 What was measured

Analysis	Purpose
Class distribution	Determine whether accuracy alone is a valid headline metric, and whether class weighting is required
Sample grid, 5 per class	Visual assessment of background clutter, subject scale, and inter-class similarity
Dimension and aspect histograms	Decide the resize and crop strategy
Perceptual duplicate audit	Quantify redundancy and prevent train/test leakage
Grouped stratified split	Produce a frozen, reusable partition for Submissions 2–4

2.2 Sample images

Figure 1 shows two representative frames from each class. They were chosen to span the range of quality present in the dataset rather than to show the clearest examples, because the variation is itself a finding.

Four observations follow directly from these samples and inform the preprocessing plan in Section 2.6.

The subject occupies a minority of the frame. In most samples the batsman covers well under half the image area. The remainder is turf, crowd, sightscreen and advertising hoardings — none of which carries class information, and all of which competes for the network's representational



Figure 1: Two samples per class. Note the shared scene grammar across all four rows: one batsman, one pitch, comparable kit and comparable backdrops. The class label is carried almost entirely by body posture and bat angle. Note also the range of capture quality — daylight and floodlit matches, net practice and match play, sharp frames and degraded broadcast captures with motion blur and colour cast.

capacity. This is the specific condition that spatial attention is intended to address in Submission 4.

Backgrounds are correlated with capture context, not with class. Advertising boards, crowd density and floodlighting vary widely but are unrelated to which shot is being played. A network that latches onto them will appear to perform well on the training distribution and fail on new footage.

Capture quality is highly variable. The dataset mixes sharp high-resolution match photography with low-resolution broadcast captures showing motion blur, compression artefacts and strong colour casts. This argues for brightness and contrast jitter during augmentation, and against any preprocessing step that assumes consistent colour balance.

Shot geometry is side-dependent. Comparing the `sweep` and `legglance-flick` rows makes clear that these classes are distinguished partly by which side of the body the ball is played to, and by the degree of knee bend. This is the direct justification for excluding horizontal flipping from the augmentation pipeline — see the note in Section 2.6.

2.3 Class balance

Quantity	Value
Largest class (name, count)	___
Smallest class (name, count)	___
Imbalance ratio	___ : 1
Verdict	___

Consequence. If the imbalance ratio exceeds roughly 1.5:1, macro-averaged F1 is reported alongside accuracy from Submission 2 onward. Above roughly 3:1, class weights are applied during training in Submission 3 and inherited unchanged by Submission 4.

2.4 Duplicate audit and leakage control

> **This is the most consequential finding of Submission 1.** The dataset was assembled by collecting images and then augmenting them offline before publication. Multiple files are therefore transformations of a single source photograph. A conventional per-file random split distributes those transformations across both the training and test partitions, so the model is evaluated on images it has effectively already seen, and *every* accuracy figure reported downstream is inflated.

2.4.1 Method

A 64-bit perceptual *difference hash* (dHash) is computed for each image. Each image is converted to greyscale, resized to 9×8 pixels, and the sign of the horizontal gradient between adjacent pixels is recorded as one bit per comparison. The resulting fingerprint is insensitive to minor changes in scale, compression and brightness, but responds to genuine differences in content. Images sharing a hash are collected into a **duplicate group**.

The implementation depends on nothing beyond Pillow and NumPy, so it runs unchanged on Kaggle, Colab or a local machine.

2.4.2 Findings

Quantity	Value
Unique perceptual hashes	___ of ___ images
Duplicate groups	___
Images inside a duplicate group	___
Redundant copies	___ (___% of dataset)
Groups spanning more than one class	___
Effective dataset size	___ independent images

Groups spanning more than one class, if any are found, represent label conflicts: perceptually identical images carrying different ground-truth labels. These are reported and inspected rather than silently removed.

2.4.3 Resulting split policy

The train/validation/test partition is constructed over *duplicate groups*, stratified by class, in a 70/15/15 ratio with seed 42. An assertion verifies that no group's members appear in more than one partition. The result is written to `artifacts/split_manifest.csv` and committed to the repository.

> **Submissions 2, 3 and 4 load this manifest. They do not re-split.** If the partition changed between Submission 3 and Submission 4, the attention comparison would be two unrelated training runs rather than a controlled experiment, and no difference between them could be attributed to the attention module.

2.5 Written findings

Complete each statement from the notebook output before submission. The reasoning is written; the numbers are not.

Finding 1 — Balance. The dataset contains ___ images across four classes, with an imbalance ratio of ___ : 1, which is ___. Accuracy alone ___ a sufficient headline metric; macro-F1 ___ reported alongside it, and class weights ___ applied.

Finding 2 — Redundancy. Perceptual hashing identified ___ duplicate groups covering ___ images, meaning ___ files (___%) are redundant. The effective dataset size is ___ independent images rather than the ___ files on disk. All partitioning is therefore performed on group identity.

Finding 3 — Class similarity. All four classes share an identical scene grammar; the discriminative signal is confined to posture and bat angle within a minority of the frame. The most visually similar pair is ___ and ___. This is the specific weakness the attention module in Submission 4 targets, by suppressing background response and concentrating representational capacity on the batsman.

Finding 4 — Geometry and noise. Images span ___–___ px in width and ___–___ px in height across ___ distinct size combinations. Backgrounds vary in illumination and crowd density, and some frames are low-resolution broadcast captures with visible compression artefacts. All images are resized to 224×224 to match the ImageNet-pretrained backbones used in Submission 2.

2.6 Preprocessing and augmentation plan

Decision	Choice	Driven by
Resize	224×224	Finding 4; pretrained backbone input size
Normalisation	ImageNet mean/std	Transfer learning in Submission 2
Horizontal flip	**Not applied**	Finding 3 — see note below
Rotation	$\pm 10°$	Finding 4; realistic broadcast camera tilt
Brightness / contrast	$\pm 20\%$	Finding 4; varying stadium illumination
Random resized crop	scale 0.8–1.0	Finding 3; forces reliance on the batsman
Class weights	—	Finding 1
Split strategy	Grouped, stratified, 70/15/15	Finding 2

On horizontal flipping. This is the augmentation most commonly applied by default in image classification pipelines, and it is incorrect for this dataset. A sweep and a leg glance are defined in part by which side of the body the ball is played to. Mirroring an image therefore changes the correct label, injecting systematic label noise rather than useful invariance. It is excluded deliberately.

3 Related work

Peer-reviewed papers and preprints only. Note that the great majority of prior work on cricket shot classification operates on *video*; this is the gap the present project addresses.

#	Paper (author, year)	Dataset	Model(s)	Approach	Result
1	Sen, Deb, Dhar & Koshiba (2021), *Sensors* — "CricShotClassify" [1]	CricShot10 (own; 10 classes, video)	Conventional CNN, dilated CNN, VGG16 / InceptionV3 / Xception / DenseNet169, each + GRU	Transfer learning with frozen backbones, then selective unfreezing of the final 4 and final 8 VGG16 layers; GRU for temporal dependency	86% (fully frozen VGG16–GRU); 93% (final 4 or 8 layers fine-tuned)
2	Kang (2025), *arXiv:2510.09187* [2]	CricShot10	Seven models: CNN+GRU, Bi-LSTM + attention, ViT, transfer learning, EfficientNet-B0 + GRU	Unified re-implementation of prior work under a single standardised evaluation protocol	92.25% (proposed). **Re-implementations of three prior published models scored 46.0%, 55.6% and 57.7% against originally reported 96%, 99.2% and 93%**
3	Dey & Biswas (2024), *Int. J. Engineering* — "Shot-ViT" [3]	CBSId (own; 2,160 annotated **images**, 7 classes)	Fine-tuned Vision Transformer, compared against VGG19, ResNet50, I-AlexNet, ViT_B32	Image-level rather than video-level classification; ViT fine-tuned for cricket batting shots	Shot-ViT outperforms all four compared CNN and ViT baselines
4	Kumar et al. (2019), *IJRTE* [4]	Own video dataset	Modified LRCN (CNN + LSTM)	CNN for spatial features, LSTM for temporal modelling	96%, but on a reduced set of **5** shot classes rather than the full 10
5	Bhat et al. (2024), *IJERCSE* [5]	Cricket shot video dataset	CNN+RNN, attention network, ViT	Direct comparison of three deep learning paradigms on one dataset	99.2% (CNN+RNN), 99.19% (attention), 98.9% (ViT) — not reproduced by the independent re-implementation in row 2
6	Foysal, Islam, Karim & Neehal (2019), *arXiv:1909.01228* [6]	BowlersNet (own; 8,100 **images**, 18 bowlers)	VGG16 + transfer learning	Single-frame image classification of cricket bowling actions; selective layer freezing	93.3% average test accuracy

3.1 What this survey establishes

1. The field is overwhelmingly video-based. Rows 1, 2, 4 and 5 all model temporal sequences. Only rows 3 and 6 operate on single images, and row 6 classifies bowlers rather than batting shots. Single-frame batting shot classification is therefore genuinely under-examined, which is the niche this project occupies.

2. Published accuracies in this area are unreliable. Kang [2] re-implemented prior models under a standardised protocol and recovered roughly half the originally reported accuracy in each case. Inconsistent data partitioning is a plausible contributor — precisely the failure mode that the duplicate audit in Section 2.4 is designed to eliminate here.

> This is why the leakage control in this project is not incidental tidiness. It is a direct response to a documented reproducibility problem in the immediate literature, and it should be cited when presenting this project's own results.

Note: citations in rows 4 and 5 of the table above are reported via the survey in [2]. Locate and verify the original publications through your institution's library before final submission.

3. Attention appears, but is never isolated. Row 5 reports an attention-based network, but introduces it alongside other architectural changes, so the contribution of attention alone cannot be separated from the rest. Submission 4 of this project holds architecture, data, split, seed and all hyperparameters constant and varies only the presence of the attention module, making the measured difference attributable.

4 Protocol for subsequent submissions

The following are fixed here and held constant throughout, so that every later comparison is attributable.

Setting	Value
Split	70 / 15 / 15, grouped by duplicate identity, stratified by class
Random seed	42
Input resolution	224×224
Normalisation	ImageNet channel statistics
Reported metrics	Accuracy, precision, recall, F1 (per class and macro), confusion matrix, training and testing time
Split source	`artifacts/split_manifest.csv` — loaded, never regenerated

Planned work

#	Deliverable	Scope
2	Pretrained models	Three ImageNet backbones (ResNet50, EfficientNet-B0, DenseNet121), each evaluated frozen and fine-tuned, 30–50 epochs — six training runs
3	Custom CNN from scratch	Four convolutional blocks with batch normalisation and ReLU, global average pooling, fully connected head; augmentation and early stopping
4	CNN with attention	The Submission 3 architecture with a CBAM module inserted after each convolutional block. Nothing else changes. Grad-CAM comparison of where each model attends

5 Submission checklist

- ☐ Dataset added as a notebook input; `DATA_DIR` resolves without error
- ☐ Licence and publication year transcribed from the Kaggle page (Section 1.1)
- ☐ Notebook executed with **Restart & Run All**, so all plots and tables are stored in the `.ipynb`
- ☐ All ____ placeholders in this report replaced with computed values
- ☐ Most-confused class pair identified from the sample grid and recorded in Finding 3
- ☐ `artifacts/split_manifest.csv` generated and committed
- ☐ Reference links verified against the original publications
- ☐ Notebook pushed to GitHub with outputs visible; repository link submitted

References

[1] A. Sen, K. Deb, P. K. Dhar, and T. Koshiba, "CricShotClassify: An Approach to Classifying Batting Shots from Cricket Videos Using a Convolutional Neural Network and Gated Recurrent Unit," *Sensors*, 2021. https://pmc.ncbi.nlm.nih.gov/articles/PMC8072636

[2] S. Kang, "Modern Deep Learning Approaches for Cricket Shot Classification: A Comprehensive Baseline Study," *arXiv:2510.09187*, 2025. https://arxiv.org/abs/2510.09187

[3] A. Dey and S. Biswas, "Shot-ViT: Cricket Batting Shots Classification with Vision Transformer Network," *International Journal of Engineering*, 2024. https://www.ije.ir/article_195819.html

[4] B. Kumar et al., "Video based cricket shot classification using modified LRCN approach," *International Journal of Recent Technology and Engineering*, vol. 8, no. 3, 2019.

[5] I. Bhat et al., "Classification of cricket shots from cricket videos using deep learning models," *International Journal of Engineering Research in Computer Science and Engineering*, vol. 11, no. 5, 2024.

[6] M. F. A. Foysal, M. S. Islam, A. Karim, and N. Neehal, "Shot-Net: A Convolutional Neural Network for Classifying Different Cricket Shots," *arXiv:1909.01228*, 2019. https://arxiv.org/abs/1909.01228